

6510 Assembly Instructions Cheat Sheet

Logical and Arithmetic

	1 byte	2 bytes	3 bytes		
	(implied)	#00 \$00 \$00,X (\$00,X) (\$00,Y) \$0000 \$0000,X \$0000,Y		* add one cycle if page boundary crossed	
ORA	09 2	05 3	15 4 6 5* 4 4* 4*	NZ	A or \$ ⇒ A
AND	29 2	25 3	35 4 6 5* 4 4* 4*	NZ	A and \$ ⇒ A
EOR	49 2	45 3	55 4 6 5* 4 4* 4*	NZ	A xor \$ ⇒ A
ADC	69 2	65 3	75 4 6 5* 4 4* 4*	NZCV	A + \$ + C ⇒ A
SBC	E9 2	E5 3	F5 4 6 5* 4 4* 4*	NZCV	A - \$ + C - 1 ⇒ A
CMP	C9 2	C5 3	D5 4 6 5* 4 4* 4*	NZC	A - \$
CPX	E0 2	E4 3	EC 4	NZC	X - \$
CPY	C0 2	C4 3	CC 4	NZC	Y - \$
DEC		C6 5	D6 6	NZ	\$ - 1 ⇒ \$
DEX	CA 2			NZ	X - 1 ⇒ X
DEY	88 2			NZ	Y - 1 ⇒ Y
INC		E6 5	F6 6	NZ	\$ + 1 ⇒ \$
INX	F8 2			NZ	X + 1 ⇒ X
INY	C8 2			NZ	Y + 1 ⇒ Y
ASL	0A 2	06 5	16 6	NZC	\$ × 2 ⇒ \$
ROL	2A 2	26 5	36 6	NZC	\$ × 2 + C ⇒ \$
LSR	4A 2	46 5	56 6	NZC	\$ / 2 ⇒ \$
ROR	6A 2	66 5	76 6	NZC	\$ / 2 + C × 128 ⇒ \$

Loads and Stores

	2 bytes	3 bytes		
	#00 \$00 \$00,X \$00,Y (\$00,X) (\$00,Y) \$0000 \$0000,X \$0000,Y		* add one cycle if page boundary crossed	
LDA	A9 2	A5 3	B5 4 6 5* 4 4* 4*	NZ \$ ⇒ A
STA		85 3	95 4 6 4 5 5	A ⇒ \$
LDX	A2 2	A6 3	B6 4	4* NZ \$ ⇒ X
STX		86 3	96 4	X ⇒ \$
LDY	A0 2	A4 3	B4 4 4*	NZ \$ ⇒ Y
STY		84 3	94 4	Y ⇒ \$

Jumps and Calls

	1 byte	3 bytes		
	(implied)	\$0000 (\$0000)	Flags	Function
BRK	7†		BI	PC + 2 ⇒ (--S), P ⇒ (--S), \$FFFE ⇒ PC
RTI	6		NZCVDI	(S++) ⇒ P, (S++) ⇒ PC
JSR	20 6			PC ⇒ (--S), \$ ⇒ PC
RTS	60 6			(S++) ⇒ PC
JMP	4C 3	6C 5		\$ ⇒ PC

† BRK pushes PC+2, padding with NOP recommended

Transfers and Stack

	1 byte		
	(implied) Flags	Function	
TAX	AA 2	NZ	A ⇒ X
TXA	8A 2	NZ	X ⇒ A
TAY	A8 2	NZ	A ⇒ Y
TYA	98 2	NZ	Y ⇒ A
TSX	BA 2	NZ	S ⇒ X
TXS	9A 2		X ⇒ S
PLA	68 4	NZ	(S++) ⇒ A
PHA	48 3		A ⇒ (--S)
PLP	28 4	NZCVDI	(S++) ⇒ P
PHP	08 3		P ⇒ (--S)

Flag Manipulation

	1 byte	2 bytes	3 bytes		
	(implied)	\$00 \$00 \$00,X \$00,Y (\$00,X) (\$00,Y) \$0000 \$0000,X \$0000,Y		Flags set \$ bit 7 ⇒ N, \$ bit 6 ⇒ V, \$ and A ⇒ Z	
BIT	24 3	2C 4			
CLC	18 2			0 ⇒ C	
SEC	38 2			1 ⇒ C	
CLD	D8 2			0 ⇒ D	
SED	F8 2			1 ⇒ D	
CLI	58 2			0 ⇒ I	
SEI	78 2			1 ⇒ I	
CLV	B8 2			0 ⇒ V	

Illegal Instructions

	2 bytes	3 bytes		
	#00 \$00 \$00,X \$00,Y (\$00,X) (\$00,Y) \$0000 \$0000,X \$0000,Y		* add one cycle if page boundary crossed	
ASO	07 5	17 6	03 8 8 0F 7 1B	NZC ASL \$, ORA \$
RLA	27 5	37 6	23 8 8 33 7 3B	NZC ROL \$, AND \$
LSE	47 5	57 6	43 8 8 4F 7 5B	NZC LSR \$, EOR \$
RRA	67 5	77 6	63 8 8 6F 7 7B	NZCV ROR \$, ADC \$
SAX	87 3	97 4	83 6 4 8F	A and X ⇒ \$
LAX	AB 2†	A7 3	B7 4 6 5* 4 4*	NZ LDA \$, LDX \$
DCM	C7 5	D7 6	C3 8 8 6F 7 7B	NZC DEC \$, CMP \$
INS	E7 5	F7 6	E3 8 8 7F 7 7B	NZCV INC \$, SBC \$
ANC	0B 2			NZC AND \$, N ⇒ C
ALR	2B 2			NZC AND \$, LSR
ARR	4B 2			NZCV AND \$, ROR
XAA	6B 2†			NZ TXA, AND \$
AXS	8B 2			NZC [A and X] - \$ ⇒ X
AXA	AB 2			A and X and H ⇒ \$
SAY	CB 2			Y and H ⇒ \$
XAS	DB 2			X and H ⇒ \$
TAS	FB 2			A and X ⇒ S, S and H
LAS	AB 2			4* NZ S and \$ ⇒ [A,X,S]

Branches

	2 bytes		
	±00	Branch on	
BPL	2* 10	N = 0	
BMI	2* 30	N = 1	
BVC	2* 50	V = 0	
BVS	2* 70	V = 1	
BCC	2* 90	C = 0	
BCS	2* B0	C = 1	
BNE	2* D0	Z = 0	
BEQ	2* F0	Z = 1	



P-register

Flags:

- bit 0 C = Carry
- bit 1 Z = Zero
- bit 2 I = Interrupt disable
- bit 3 D = Decimal
- bit 4 B = Break (BRK hit)
- bit 6 V = Overflow (sign changed)
- bit 7 N = Negative (bit 7 set)

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version 0.1.0. Sources:
<http://www.oxron.de/html/opcodes02.html>
<http://unusedino.de/ec64/technical/aay/c64/>
http://codebase64.org/doku.php?id=base:6502_6510_coding

* add one cycle if branch is taken,
another if page boundary crossed

† highly unstable ‡ partially unstable H = high byte of the address + 1